

UNIT-1:

MATRICES

Introduction:

Matrix: Rectangular arrangement of elements in rows and columns is known as a matrix.

Ex:

$$\begin{bmatrix} 0 & 1 & 6 \\ 7 & 8 & 4 \\ 3 & 6 & 9 \end{bmatrix}$$

→ Matrix is generally denoted by capital letters and the elements by small letters i.e. ex: $A = [a_{ij}]_{m \times n}$

$a_{ij} \rightarrow$ 'a' element in i^{th} row and j^{th} column

Ex: $A = \begin{bmatrix} 2 & 3 & 4 \\ 6 & 7 & 9 \end{bmatrix}_{2 \times 3}$

$$a_{11} \rightarrow 2$$

$$a_{21} \rightarrow 6$$

$$a_{12} \rightarrow 3$$

$$a_{22} \rightarrow 7$$

$$a_{13} \rightarrow 4$$

$$a_{23} \rightarrow 9$$

order of the matrix:

The order of the matrix is given by m by n where ' m ' denotes the number of rows and ' n ' denotes the number of columns.

square matrix;

If number of rows is equal to number of columns then it is a square matrix i.e. $m=n$ otherwise rectangular matrix.

Row matrix;

A matrix with one row is known as a row matrix whose order is $1 \times n$

Ex: $[1 \ 3 \ 5]_{1 \times 3}$

Column matrix; A matrix with one column is known as column matrix whose order is $n \times 1$

Ex:

$$\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}_{3 \times 1}$$

Diagonal matrix:

A matrix with all elements zero except the principal diagonal square

is known as diagonal matrix. i.e. $a_{ij} = 0 ; i \neq j$

Ex:
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}_{3 \times 3}$$

scalar matrix:

A diagonal matrix with same principal diagonal elements is known as scalar matrix. $a_{ij} = 0 ; i \neq j ; = K, i = j$

Ex:
$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

unit matrix (or) Identity matrix:

A scalar matrix with diagonal element is 1 is known as unit matrix i.e. $a_{ij} = 1 ; i = j$

$a_{ij} = 0 ; i \neq j$

Ex:
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

singular matrix: A square matrix A is said to be singular if determinant (or) det of A is equal to zero otherwise it is said to be non singular matrix.

Inverse of a matrix: The square matrix B of order n is said to be inverse of a square matrix A of same order if $AB = BA = I$

Transpose of a matrix: The matrix obtained by interchanging rows and columns of the given matrix is known as transpose of matrix.

Ex:
$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3 \times 3}$$

$$A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \end{bmatrix}_{2 \times 3}$$

$$A^T = \begin{bmatrix} 2 & 1 \\ 4 & 3 \\ 6 & 5 \end{bmatrix}_{3 \times 2}$$

symmetric matrix: A square matrix is said to be symmetric if

$$A = A^T \text{ i.e. } a_{ij} = a_{ji}$$

Ex: $A = \begin{bmatrix} 2 & 4 & 5 \\ 4 & 6 & 7 \\ 5 & 7 & 8 \end{bmatrix}$ $A^T = \begin{bmatrix} 2 & 4 & 5 \\ 4 & 6 & 7 \\ 5 & 7 & 8 \end{bmatrix}$

skew-symmetric matrix: A square matrix is said to be

skew-symmetric if $A = -A^T$ i.e. $a_{ij} = -a_{ji}$

→ The diagonal elements of a skew symmetric matrix is equal to zero

$$a_{ij} = -a_{ji}$$

Ex: $A = \begin{bmatrix} 0 & 2 & -5 \\ -2 & 0 & 9 \\ 5 & -7 & 0 \end{bmatrix}$

$$B = \begin{bmatrix} 0 & -8 & -9 \\ 8 & 0 & -6 \\ 9 & 6 & 0 \end{bmatrix}$$

complex matrix: A matrix with atleast one complex number $a+ib$ is known as a complex matrix.

Ex: $A = \begin{bmatrix} 2 & 4 & 6i \\ 5 & 8 & 9 \\ 6 & 7i & 4 \end{bmatrix}$

conjugate of a matrix: The matrix obtained by taking the conjugates of elements of the given matrix is known as conjugate matrix denoted by $\bar{A}$

$$\bar{A} = \begin{bmatrix} 2 & 4 & -6i \\ 5 & 8 & 9 \\ 6 & -7i & 4 \end{bmatrix}$$

→ conjugate of real number is itself

Ex: conjugate of 3 is 3

conjugate transpose of a matrix: The transpose of a conjugated matrix is known as conjugate transpose of a matrix denoted by

$$A^0 \text{ (or) } \bar{A}^T \quad A^0 = \begin{bmatrix} 2 & 5 & 6 \\ 4 & 8 & -7i \\ -6i & 9 & 4 \end{bmatrix}$$

Hermitian matrix: A square matrix is said to be hermitian matrix if $A = A^T$ i.e. $a_{ij} = \bar{a}_{ji}$

→ The diagonal elements of a hermitian matrix are real

$$A = \begin{bmatrix} 3 & -4i & 8 \\ 4i & 6 & 3+8i \\ 8 & 3-8i & 9 \end{bmatrix}$$

skew-hermitian matrix: A square matrix is said to be skew-hermitian matrix if $A = -A^T$ i.e. $a_{ij} = -\bar{a}_{ji}$

NOTE:

The diagonal elements of a skew hermitian matrix are either zero (or) purely imaginary.

$$\begin{bmatrix} 7i & 6 & 8i \\ -6 & 9i & -2 \\ 8i & 2 & 0 \end{bmatrix}$$

Elementary transformation:

In the elementary transformations

a) row transformations:

→ We can interchange any two rows R_i & R_j denoted by $R_i \leftrightarrow R_j$

→ Multiply any row with a non-zero scalar denoted by

$$R_i \rightarrow kR_i$$

→ We can multiply any row with a non zero number and can be added to the corresponding elements of another row, denoted as $R_i \rightarrow R_i + R_j$ (or) $R_i \rightarrow kR_i \pm R_j$

b) column transformations:

→ we can interchange any two columns C_i & C_j denoted by $C_i \leftrightarrow C_j$

→ Multiply any column with a non zero scalar denoted by $C_i \rightarrow k \cdot C_i$

→ we can multiply any column with a non-zero number and can be added to the corresponding elements of another column denoted as $C_i \rightarrow C_i + k(C_j)$ (or) $C_i \rightarrow kC_i \pm C_j$

Rank of the matrix:

Rank: A positive integer "r" is said to be the rank of the matrix A of order $m \times n$ if

a) Every minor of order $r+1$ is zero and

b) At least one minor of order r is not equal to zero

NOTE:

'A' is a matrix of order $m \times n$ then rank is less than or equal to minimum of m, n [$r \leq \min(m, n)$]

→ The rank of null matrix is zero

→ The rank of any matrix is never equal to zero except for null matrix.

→ The rank of a non singular matrix of order n is n

→ The rank of Identity matrix of order n is n

→ Every matrix will have a rank and it's unique

→ Rank is also denoted by $\rho(\text{row})$

→ Rank of $A = \bar{A}^T$

$$[\rho(A) = \rho(A)^T]$$

→ Elementary transformation does not change the rank

Rank by Echelon form:

Echelon form:

A matrix is said to be in echelon form if

- The first element of the matrix is non zero.
- The no. of zeroes before the first non-zero element is less than no. of such zeroes in the next row.

Note:

The number of non-zero rows in the echelon form gives the rank

NOTE:

only row-operations should be applied to reduce the matrix into echelon form.

Q) Reduce the matrix into the echelon form and find the rank.

$$\begin{bmatrix} 1 & 2 & -4 & 5 \\ 2 & -1 & 3 & 6 \\ 8 & 1 & 9 & 7 \end{bmatrix}_{3 \times 4}$$

sol) $\begin{bmatrix} 1 & 2 & -4 & 5 \\ 2 & -1 & 3 & 6 \\ 8 & 1 & 9 & 7 \end{bmatrix}$

$$R_2 \rightarrow R_2 - 2R_1 ; R_3 \rightarrow R_3 - 8R_1$$

$$\begin{bmatrix} 1 & 2 & -4 & 5 \\ 0 & -5 & 11 & -4 \\ 0 & -15 & 41 & -33 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2$$

$$\begin{bmatrix} 1 & 2 & -4 & 5 \\ 0 & -5 & 11 & -4 \\ 0 & 0 & 8 & -21 \end{bmatrix}$$

Rank = 3

Q) Reduce the matrix into the echelon form and find the rank.

$$\begin{bmatrix} 2 & 1 & 3 & 5 \\ 4 & 2 & 1 & 3 \\ 8 & 4 & 7 & 13 \\ 8 & 4 & -3 & -1 \end{bmatrix}$$

Sol)

$$\begin{bmatrix} 2 & 1 & 3 & 5 \\ 4 & 2 & 1 & 3 \\ 8 & 4 & 7 & 13 \\ 8 & 4 & -3 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1; R_3 \rightarrow R_3 - 4R_1; R_4 \rightarrow R_4 - 4R_1$$

$$\begin{bmatrix} 2 & 1 & 3 & 5 \\ 0 & 0 & -5 & -7 \\ 0 & 0 & -5 & -7 \\ 0 & 0 & -15 & -21 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_3; R_3 \rightarrow R_3$$

$$R_3 \rightarrow R_3 - R_2; R_4 \rightarrow R_4 - 3R_2$$

$$\begin{bmatrix} 2 & 1 & 3 & 5 \\ 0 & 0 & -5 & -7 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank} = 2$$

Rank by Normal form;

→ A matrix is said to be in normal form if it is in any one of the following forms,

$$I_1; [I_1 \ 0]; \begin{bmatrix} I_1 \\ 0 \end{bmatrix}; \begin{bmatrix} I_1 & 0 \\ 0 & 0 \end{bmatrix}$$

NOTE:

→ To reduce the matrix to the normal form we use both row and column transformations.

Q) Find the rank of matrix by reducing to the normal form.

$$\begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{bmatrix}$$

$$R_1 \times (-1)$$

$$\begin{bmatrix} 1 & 3 & -3 & 1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1; R_3 \rightarrow R_3 - 2R_1; R_4 \rightarrow R_4 + R_1$$

$$\begin{bmatrix} 1 & 3 & -3 & 1 \\ 0 & -2 & 2 & -1 \\ 0 & -11 & 8 & -5 \\ 0 & 4 & -3 & 2 \end{bmatrix}$$

$$C_2 \rightarrow C_2 - 3C_1; C_3 \rightarrow C_3 + 3C_1; C_4 \rightarrow C_4 - C_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 2 & -1 \\ 0 & -11 & 8 & -5 \\ 0 & 4 & -3 & 2 \end{bmatrix}$$

$$R_3 \rightarrow 2R_3 - 11R_2; R_4 \rightarrow R_4 + 2R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 2 & -1 \\ 0 & 0 & -6 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$C_3 \rightarrow C_3 + C_2; C_4 \rightarrow 2C_4 - C_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$R_4 \rightarrow 6R_4 + R_3$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$\frac{C_2}{-2}, \frac{C_3}{-6}, \frac{C_4}{2}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C_4 \rightarrow C_4 - C_3$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rank of matrix is 4

Inverse by Gauss-Jordan Method:

Inverse: A non-singular matrix A of order n is said to be inverse of another non-singular matrix B of same order if $AB = BA = I$

Procedure:

Step 1: Write the given matrix A

$$A = IA$$

Step 2: Apply the row transformations on A in LHS so that it reduces to identity matrix. The same row transformations were applied on I in RHS then it reduces to the form

$$I = A^{-1}A$$

Q. Find the inverse of

$$\begin{bmatrix} 4 & -1 & 1 \\ 2 & 0 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

sol

$$A = IA$$

$$\begin{bmatrix} 4 & -1 & 1 \\ 2 & 0 & -1 \\ 1 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & -1 \\ 4 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 - 2R_1; R_3 \rightarrow R_3 - 4R_1$$

$$\begin{bmatrix} 1 & -1 & 3 \\ 0 & 2 & -7 \\ 0 & 3 & -11 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -2 \\ 1 & 0 & -4 \end{bmatrix} A$$

$$R_1 \rightarrow 2R_1 + R_2; R_3 \rightarrow 2R_3 - 3R_2$$

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & -7 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & -2 \\ 2 & -3 & -2 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 - R_3; R_2 \rightarrow R_2 - 7R_3$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 4 & 2 \\ -14 & 22 & 12 \\ 2 & -3 & -2 \end{bmatrix} A$$

$$\frac{R_1}{2}, \frac{R_2}{2}, R_3 (-1)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 1 \\ -7 & 11 & 6 \\ -2 & 3 & 2 \end{bmatrix} A$$

$$A^{-1} = \begin{bmatrix} -1 & 2 & 1 \\ -7 & 11 & 6 \\ -2 & 3 & 2 \end{bmatrix}$$

solution of linear system of equations:

The set of equation

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_n$$

which are m equations in ' n ' unknown is known as linear system of equations whose matrix form is given $AX=B$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}$$

known as coefficient matrix

→ $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}$ known as matrix of unknowns

→ $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}_{m \times 1}$ known as constant matrix

→ If $b=0$ (i.e. RHS=0) then the system is known as homogeneous otherwise it is known as non-homogeneous system.

solution of non-homogeneous system of equations:

→ If $b \neq 0$ then the system $AX=B$ is known as non-homogeneous

Procedure:

step 1: consider argument matrix $C = [A/B]$

step 2: Reduce C to the echelon form.

step 3: a) If $\rho(A) \neq \rho(C)$ then the system will have no solution and is said to be inconsistent.

⊙ If $\rho(A) = \rho(C) = n$ [n = number of unknowns (or) variables] then the system is said to be consistent and will have unique solution

⊙ If $\rho(A) = \rho(C) < n$ then the system is said to be consistent and will have infinite number of solutions, which are obtained by assigning $n-R$ variable and arbitrary value.

Q) solve $x+y+2z=4$, $2x-y+3z=9$, $3x-y-z=2$

sol) The matrix form of a given equations is $AX=B$

where $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 3 \\ 3 & -1 & -1 \end{bmatrix}$ $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $B = \begin{bmatrix} 4 \\ 9 \\ 2 \end{bmatrix}$

consider argument matrix $C = [A/B]$

$$C = [A/B] = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 2 & -1 & 3 & 9 \\ 3 & -1 & -1 & 2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 ; R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & -3 & -1 & 1 \\ 0 & -4 & -7 & -10 \end{bmatrix}$$

$$R_3 \rightarrow 3R_3 - 4R_2$$

$$\begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & -3 & -1 & 1 \\ 0 & 0 & -12 & -34 \end{bmatrix}$$

Here $\rho(A) = \rho(C) = 3 = n$

The system will have unique solution which is given by back substitution.

$$0x + 0y - 12z = -34 \text{ (last row)}$$

$$\boxed{z = 2}$$

$$6x - 3y - z = 1 \text{ (2nd row)}$$

$$-3y - z = 1$$

$$\boxed{y = -1}$$

$$x + y + 2z = 4 \text{ (1st row)}$$

$$x - 1 + 4 = 4$$

$$\boxed{x = 1}$$

Q) solve $x - 4y + 7z = 14, 3x + 8y - 2z = 13, 7x - 8y + 26z = 5$

soln

$$A = \begin{bmatrix} 1 & -4 & 7 \\ 3 & 8 & -2 \\ 7 & -8 & 26 \end{bmatrix} \quad x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} 14 \\ 13 \\ 5 \end{bmatrix}$$

consider augmented matrix $C = [A|B]$

$$C = [A|B] = \begin{bmatrix} 1 & -4 & 7 & 14 \\ 3 & 8 & -2 & 13 \\ 7 & -8 & 26 & 5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$R_3 \rightarrow R_3 - 7R_1$$

$$= \begin{bmatrix} 1 & -4 & 7 & 14 \\ 0 & 20 & -23 & -29 \\ 0 & 20 & -23 & -93 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & -4 & 7 & 14 \\ 0 & 20 & -23 & -29 \\ 0 & 0 & 0 & -64 \end{bmatrix}$$

$$\text{Here } \rho(A) \neq \rho(C) = 3$$

$$\rho(A) = 2, \rho(C) = 3$$

The system is inconsistent (i.e. no solution)

Q) solve $x - y + 2z = 4$; $3x + y + 4z = 6$; $x + y + z = 1$

Sol) The matrix of given equations is in the form $AX = B$

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 1 & 4 \\ 1 & 1 & 1 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} 4 \\ 6 \\ 1 \end{bmatrix}$$

augmented matrix $C[A/B]$

$$C = \begin{bmatrix} 1 & -1 & 2 & 4 \\ 3 & 1 & 4 & 6 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1; R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 4 & -2 & -6 \\ 0 & 2 & -1 & -3 \end{bmatrix}$$

$$R_3 \rightarrow 2R_3 - R_2$$

$$\begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 4 & -2 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Here } \rho(A) = \rho(C) = 2$$

$$n = 3$$

$$\text{rank} < n$$

The system will have infinite numbers of solutions

$$\text{Here } n - b = 3 - 2 = 1$$

1 variable must assign arbitrary value

$$\text{let } \boxed{z = k}$$

$$4y - 2z = -6$$

$$2y - z = -3$$

$$2y = -3 + k$$

$$\boxed{y = \frac{-3+k}{2}}$$

$$x - y + 2z = 4$$

$$x - \left[\frac{-3+k}{2} \right] + 2z = 4$$

$$x - \frac{3-k}{2} + 2z = 4$$

$$2x - 3 - k + 4k = 8$$

$$2x = 5 - 3k$$

$$\boxed{x = \frac{5-3k}{2}}$$

Q) Find the values of λ & μ the system of equations
 $x+y+z=6$; $x+2y+3z=10$; $x+2y+\lambda z=\mu$ have
 (a) No solution (b) unique solution (c) infinite solution.

sol)

$$C[A/B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1; R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{bmatrix}$$

(a) No solution:

$$\rho(A) \neq \rho(C)$$

$$\lambda - 3 = 0$$

$$\lambda = 3$$

$$\mu - 10 \neq 0$$

$$\mu \neq 10$$

(b) unique solution:

$$\rho(A) = \rho(C)$$

$$\lambda - 3 \neq 0$$

$$\lambda \neq 3$$

$$\mu - 10 = 0$$

$$\mu = 10$$

(or)

μ can be any value

(c) infinite solution:

$$\rho(A) = \rho(C) < 3$$

$$\lambda - 3 = 0, \mu - 10 = 0$$

$$\boxed{\lambda = 3} \quad \boxed{\mu = 10}$$

Q) Find the value of λ for which the system of equations $3x - y + 4z = 3$, $x + 2y - 3z = -2$, $6x + 5y + \lambda z = -3$ will have infinite number of solutions and also solve them.

Sol

$$C[A/B] = \begin{bmatrix} 3 & -1 & 4 & 3 \\ 1 & 2 & -3 & -2 \\ 6 & 5 & \lambda & -3 \end{bmatrix}$$

$$R_2 \leftrightarrow R_1$$

$$\begin{bmatrix} 1 & 2 & -3 & -2 \\ 3 & -1 & 4 & 3 \\ 6 & 5 & \lambda & -3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1; R_3 \rightarrow R_3 - 6R_1$$

$$\begin{bmatrix} 1 & 2 & -3 & -2 \\ 0 & -7 & 13 & 9 \\ 0 & -7 & \lambda+18 & 9 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & -3 & -2 \\ 0 & -7 & 13 & 9 \\ 0 & 0 & \lambda+5 & 0 \end{bmatrix}$$

The equations have infinite solutions

$$\rho(A) = \rho(C) < 3$$

$$\text{Rank} = 2$$

$$\text{POB } \lambda+5=0$$

$$\boxed{\lambda = -5}$$

$$\rho(A) = \rho(C) < n$$

$$n - \rho = 3 - 2 = 1$$

$$\boxed{z = k}$$

$$-7y + 13z = 9$$

$$-7y + 13k = 9$$

$$\boxed{y = \frac{13k-9}{7}}$$

$$x + 2y - 3z = -2$$

$$x + 2\left[\frac{13k-9}{7}\right] - 3k = -2$$

$$x + \frac{26k-18}{7} - 3k = -2$$

$$7x + 26k - 18 - 21k = -14$$

$$7x = -5k + 18 - 14$$

$$7x = -5k + 4$$

$$x = \frac{4 - 5k}{7}$$

solution of homogeneous system of Equations:

If $B=0$ (i.e. $AX=0$) is known as homogeneous system of equation

Here coefficient matrix is equal to argument matrix

$$\delta(A) = \delta(C)$$

case i: If $\delta(A) = n$ (no of variables) then the system will have unique solution which is $x=0$ known as trivial solution.

case ii: If $\delta(A) < n$ then the system will have infinite number of solutions known as non-trivial solution

which are obtained by assigning $n - \delta$ variables arbitrary value i.e. system will have $n - \delta$ independent solutions

Q) solve $x_1 + 2x_3 - 2x_4 = 0$; $2x_1 - x_2 - x_4 = 0$; $x_1 + 2x_3 - x_4 = 0$;
 $4x_1 - x_2 + 3x_3 - x_4 = 0$

$$\begin{bmatrix} 1 & 0 & 2 & -2 \\ 2 & -1 & 0 & -1 \\ 1 & 0 & 2 & -1 \\ 4 & -1 & 3 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1; R_3 \rightarrow R_3 - R_1; R_4 \rightarrow R_4 - 4R_1$$

$$\begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & -1 & -4 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & -1 & -5 & 7 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - R_2$$

$$\begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & -1 & -4 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 4 \end{bmatrix}$$

$$R_3 \leftrightarrow R_4$$

$$\begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & -1 & -4 & 3 \\ 0 & 0 & -1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\rho(A) = 4 = n$$

system will have unique solution which is a trivial solution given by $x = 0$ (i.e. $x_1 = x_2 = x_3 = x_4 = 0$)

a) solve $x + 3y - 2z = 0$; $2x - y + 4z = 0$; $x - 11y + 14z = 0$

sol) $\begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix}$

$$R_2 \rightarrow R_2 - 2R_1; R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & -14 & 16 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Here } \rho(A) = 2 < n$$

system will have non trivial solution (i.e. infinite number of solutions)

$$[n - \rho = 3 - 2 = 1] \text{ one independent solution}$$

Here $x + 3y - 2z = 0 \rightarrow \textcircled{1}$

$$-7y + 8z = 0 \rightarrow \textcircled{2}$$

$$\text{let } \boxed{z = k}$$

$$\textcircled{2} \Rightarrow -7y + 8k = 0$$

$$-7y = -8k$$

$$\boxed{y = \frac{8}{7}k}$$

$$\textcircled{1} \Rightarrow x + 3\left(\frac{8}{7}k\right) - 2k = 0$$

$$7x + 24k - 14k = 0$$

$$7x = -10k$$

$$\boxed{x = -\frac{10}{7}k}$$

$$x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -\frac{10}{7}k \\ \frac{8}{7}k \\ k \end{bmatrix} = \frac{k}{7} \begin{bmatrix} -10 \\ 8 \\ 7 \end{bmatrix}$$

Q) solve $4x + 2y + z + 3w = 0$; $6x + 3y + 4z + 7w = 0$; $2x + y + w = 0$

Sol

$$A = \begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow 4R_2 - 6R_1; R_3 \rightarrow 4R_3 - 2R_1$$

$$\begin{bmatrix} 4 & 2 & 1 & 3 \\ 0 & 0 & 10 & 10 \\ 0 & 0 & -2 & -2 \end{bmatrix}$$

$$R_3 \rightarrow 5R_3 + R_2$$

$$\begin{bmatrix} 4 & 2 & 1 & 3 \\ 0 & 0 & 10 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2 / 10$$

$$\begin{bmatrix} 4 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Here $\rho(A) = 2$ and $n = 4$

The system will have $n - \rho = 4 - 2 = 2$ solution have 2 independent solution

$$4x + 2y + z + 3w = 0 \rightarrow \textcircled{1}$$

$$z + w = 0$$

$$\boxed{z = -w} \rightarrow \textcircled{2}$$

let $w = k_1$ $z = -k_1$ and $y = k_2$

$$4x + 2y + z + 3w = 0$$

$$4x + 2k_2 - k_1 + 3k_1 = 0$$

$$4x + 2k_2 + 2k_1 = 0$$

$$2x + k_2 + k_1 = 0$$

$$x = \frac{-k_1 - k_2}{2}$$

$$X = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -\frac{k_1}{2} - \frac{k_2}{2} \\ k_2 \\ -k_1 \\ k_1 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{k_1}{2} \\ 0 \\ -k_1 \\ k_1 \end{bmatrix} + \begin{bmatrix} -\frac{k_2}{2} \\ k_2 \\ 0 \\ 0 \end{bmatrix}$$

$$\frac{k_1}{2} \begin{bmatrix} -1 \\ 0 \\ -2 \\ 2 \end{bmatrix} + \frac{k_2}{2} \begin{bmatrix} -1 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

Q Find the value of k if the rank is 2

$$A = \begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & k & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1; R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & k+1 & -2 \\ 0 & -2 & 3 & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & k+1 & -2 \\ 0 & 0 & 2-k & 0 \end{bmatrix}$$

As the rank is 2

$$2 - k = 0$$

$$\boxed{k = 2}$$

Q3 Show that the only real number λ for which the system of equations
 $x + 2y + 3z = \lambda x$, $3x + y + 2z = \lambda y$, $2x + 3y + z = \lambda z$ has a non-zero solution
 if $\lambda = 6$

Sol For the homogeneous system to have non trivial (non-zero) solution,
 $\rho(A) < n$.

$$\det A = 0$$

$$|A| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 & 3 \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{vmatrix} = 0$$

$$= 1-\lambda ((1-\lambda)^2 - 6) - 2(3 - 3\lambda - 4) + 3(9 - [2(1-\lambda)])$$

$$= (1-\lambda)[1+\lambda^2 - 2\lambda - 6] - 2(-3\lambda - 1) + 3(9 - 2 + 2\lambda)$$

$$= 1 + \lambda^2 - 2\lambda - 6 - \lambda - \lambda^3 + 2\lambda^2 + 6\lambda + 6\lambda + 2 + 2\lambda - 6 + 6\lambda$$

$$= -\lambda^3 + 3\lambda^2 + 15\lambda + 8 = 0$$

$$\lambda = 6, \frac{-3 + \sqrt{3}i}{2}, \frac{-3 - 5i}{2}$$

$\therefore \lambda = 6$ is the only real root for which the system will have non trivial solution.

For $\lambda = 6$

$$A = \begin{bmatrix} -5 & 2 & 3 \\ 3 & -5 & 2 \\ 2 & 3 & -5 \end{bmatrix}$$

$$R_2 \rightarrow 5R_2 + 3R_1, R_3 \rightarrow 5R_3 + 2R_1,$$

$$\begin{bmatrix} -5 & 2 & 3 \\ 0 & -19 & 19 \\ 0 & 19 & -19 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\begin{bmatrix} -5 & 2 & 3 \\ 0 & -19 & 19 \\ 0 & 0 & 0 \end{bmatrix}$$

$$-5x + 2y + 3z = 0 \rightarrow \textcircled{1}$$

$$-19y + 19z = 0$$

$$-y + z = 0$$

$$\boxed{z = y}$$

$$\text{let } z = k, y = k$$

$$\text{from } \textcircled{1} \quad -5x + 2k + 3k = 0$$

$$+ 5x = -5k$$

$$\boxed{x = k}$$

Gauss elimination method / pivoting method:

Q) solve $3x + y - z = 3$; $2x - 8y + z = -5$; $x - 2y + 9z = 8$ by

Gauss elimination method

Sol)

$$A = \begin{bmatrix} 3 & 1 & -1 & 3 \\ 2 & -8 & 1 & -5 \\ 1 & -2 & 9 & 8 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - \frac{2}{3}R_1; R_3 \rightarrow R_3 - \frac{1}{3}R_1$$

$$\begin{bmatrix} 3 & 1 & -1 & 3 \\ 0 & -26/3 & 5/3 & -7 \\ 0 & -7/5 & 28/3 & 7 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{7}{26}R_2$$

$$\begin{bmatrix} 3 & 1 & -1 & 3 \\ 0 & -26/3 & 5/3 & -7 \\ 0 & 0 & \frac{231}{26} & \frac{231}{26} \end{bmatrix}$$

Since $\rho(A) = \rho(C) = 3$

system will have unique solution (which is obtained by back substitution)

$$\frac{231}{26} z = \frac{231}{26}$$

$$\boxed{z=1}$$

$$-\frac{26}{3}y + \frac{5}{3}z = -7$$

$$\frac{-26y+5}{3} = -7$$

$$-26y+5 = -21$$

$$-26y = -21-5$$

$$-26y = -26$$

$$\boxed{y=1}$$

$$3x + y - z = 3$$

$$3x + 1 - 1 = 3$$

$$\boxed{x=1}$$

Linear dependent and independent vectors

→ Two vectors x_1, x_2 are said to be linearly dependent if there exist two scalars k_1 & k_2 not both zero such that

$$k_1 x_1 + k_2 x_2 = 0$$

→ Two vectors x_1 and x_2 are said to be linearly independent if $k_1 x_1 + k_2 x_2 = 0$ such that both k_1 & k_2 are zero.

Note:

→ If the rank of the matrix of given vectors is equal to number of vectors then the vectors are linearly independent

other wise linearly dependent

a) check whether the vectors $(3,1,1)$ $(2,0,-1)$ $(4,2,1)$ are linearly dependent (or) independent.

sol) $A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 0 & -1 \\ 4 & 2 & 1 \end{bmatrix}$

$$R_2 \rightarrow 3R_2 - 2R_1, R_3 \rightarrow 3R_3 - 4R_1$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 0 & -2 & -5 \\ 0 & 2 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 0 & -2 & -5 \\ 0 & 0 & -6 \end{bmatrix}$$

Here rank = 3 = number of vectors
vectors are linearly independent.

q) check whether the given vectors $(3,1,-4)$ $(2,2,-3)$ $(0,-4,1)$ are linearly dependent (or) independent and find the relation between them.

sol) $\begin{bmatrix} 3 & 1 & -4 \\ 2 & 2 & -3 \\ 0 & 4 & 1 \end{bmatrix}$

$$R_2 \rightarrow 3R_2 - 2R_1$$

$$\begin{bmatrix} 3 & 1 & -4 \\ 0 & 4 & -1 \\ 0 & -4 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\begin{bmatrix} 3 & 1 & -4 \\ 0 & 4 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank = 2 < number of vectors

The vectors are linearly dependent

Here $3\lambda_1 + \lambda_2 - 4\lambda_3 = 0 \rightarrow (1)$

$$4\lambda_2 - \lambda_3 = 0 \rightarrow (2)$$

let $\lambda_3 = k$

$$4\lambda_2 - k = 0$$

$$4\lambda_2 = k$$

$$\boxed{\lambda_2 = \frac{k}{4}}$$

$$(1) \Rightarrow 3\lambda_1 + \lambda_2 - 4\lambda_3 = 0$$

$$3\lambda_1 + \frac{k}{4} - 4k = 0$$

$$12\lambda_1 + k - 16k = 0$$

$$12\lambda_1 = 15k$$

$$\boxed{\lambda_1 = \frac{5}{4}k}$$

since the three vectors x_1, x_2, x_3 are dependent we have $\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 = 0$ such that $\lambda_1, \lambda_2, \lambda_3$ not all zero

$$\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 = 0$$

$$\frac{5}{4}k x_1 + \frac{k}{4} x_2 + k x_3 = 0$$

$$5x_1 + x_2 + 4x_3 = 0$$

verification:

$$5 \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ -4 \end{bmatrix} + 4 \begin{bmatrix} -4 \\ -3 \\ 1 \end{bmatrix} = 0$$

Gauss seidel Method:

Diagonally dominant:

A matrix is said to be diagonally dominant if the diagonal is non-zero and larger than remaining elements

Q) using gauss seidel method solve $8x_1 - 3x_2 + 2x_3 = 20$; $4x_1 + 11x_2 - x_3 = 33$,
 $6x_1 + 3x_2 + 12x_3 = 36$

Sol) $A = \begin{bmatrix} 8 & -3 & 2 \\ 4 & 11 & -1 \\ 6 & 3 & 12 \end{bmatrix}$

As A is diagonally dominant matrix we can apply Gauss seidel method

method: $8x_1 - 3x_2 + 2x_3 = 20 \rightarrow (1)$

$4x_1 + 11x_2 - x_3 = 33 \rightarrow (2)$

$6x_1 + 3x_2 + 12x_3 = 36 \rightarrow (3)$

$x_1 = \frac{1}{8} [20 + 3x_2 - 2x_3] \rightarrow (a)$

$x_2 = \frac{1}{11} [33 - 4x_1 - x_3] \rightarrow (b)$

$x_3 = \frac{1}{12} [36 - 6x_1 - 3x_2] \rightarrow (c)$

1st iteration:

a) Take $x_2 = 0, x_3 = 0$ in (a)

$x_1^{(1)} = \frac{1}{8} [20 + 0 - 0] = 2.5$

b) put $x_1 = 2.5, x_3 = 0$ in (b)

$x_2^{(1)} = \frac{1}{11} [33 - 4(2.5) - 0] = 2.0909$

c) put $x_1 = 2.5, x_2 = 2.0909$ in (c)

$x_3^{(1)} = \frac{1}{12} [36 - 6(2.5) - 3(2.0909)] = 1.2272$

$x_1^{(1)} = 2.5, x_2^{(1)} = 2.0909, x_3^{(1)} = 1.2272$

2nd Iteration:

a) Take $x_2 = 2.0909$; $x_3 = 1.2272$ in (a)

$$x_1^{(2)} = \frac{1}{8} [20 + 3(2.0909) - 2(1.2272)] = 2.9772$$

b) Take $x_1 = 2.9772$; $x_3 = 1.2272$ in (b)

$$x_2^{(2)} = \frac{1}{11} [33 - 4(2.9772) + 1.2272] = 2.0289$$

c) Take $x_1 = 2.9772$, $x_2 = 2.0289$

$$x_3^{(2)} = \frac{1}{12} [36 - 6(2.9772) - 3(2.0289)] = 1.0041$$

3rd Iteration:

a) Take $x_2 = 2.0289$, $x_3 = 1.0041$

$$x_1^{(3)} = \frac{1}{8} [20 + 3(2.0289) - 2(1.0041)] = 3.0098$$

b) Take $x_1 = 3.0098$, $x_3 = 1.0041$

$$x_2^{(3)} = \frac{1}{11} [33 - 4(3.0098) + 1.0041] = 1.9968$$

c) Take $x_2 = 1.9968$, $x_1 = 3.0098$.

$$x_3^{(3)} = \frac{1}{12} [36 - 6(3.0098) - 3(1.9968)] = 0.9959$$

4th iteration:

a) Take $x_2 = 1.9968$, $x_3 = 0.9959$

$$x_1^{(4)} = \frac{1}{8} [20 + 3(1.9968) - 2(0.9959)] = 2.9998$$

b) Take $x_1 = 2.9998$, $x_3 = 0.9959$

$$x_2^{(4)} = \frac{1}{11} [33 - 4(2.9998) + 0.9959] = 1.9997$$

c) Take $x_1 = 2.9998$, $x_2 = 1.9997$

$$x_3^{(4)} = \frac{1}{12} [36 - 6(2.9998) - 3(1.9997)] = 1.0001$$

5th iteration:

a) Take $x_2 = 1.9997$, $x_3 = 1.0001$.

$$x_1^{(5)} = \frac{1}{8} [20 + 3(1.9997) - 2(1.0001)] = 2.9998$$

b) Take $x_1 = 2.9998$, $x_3 = 1.0001$

$$x_2^{(5)} = \frac{1}{11} [33 - 4(2.9998) + 1.0001] = 2.00008$$

c) Take $x_1 = 2.9998$, $x_2 = 2.00008$

$$x_3^{(5)} = \frac{1}{12} [36 - 6(2.9998) - 3(2.00008)] = 1.0001$$